

- ① The radius of curvature of a convex mirror is 30cm. It forms an image of an object which is half the size of the object. Find the separation between the object and the image.
- ② A galvanometer shows full scale deflection for a current I_g . If a shunt resistance S_1 is connected to the galvanometer, it gets converted into an ammeter of range $(0-I)$. When a shunt resistance of S_2 is connected the range becomes $(0-2I)$. Then $\frac{S_1}{S_2}$ is ?
- ③ Two long straight thin wires AB and CD have linear charge densities $10 \mu\text{C/m}$ and $-20 \mu\text{C/m}$ respectively. They are kept parallel to each other at a distance of 1m. Find the magnitude and direction of the net electric field at a point midway between them.
- ④ The critical angle for glass is θ_1 and that for water is θ_2 . The critical angle for glass-water surface would be (Given $\mu_{\text{fg}} = 1.5$, $\mu_{\text{fw}} = 1.33$)
a) less than θ_2 b) between θ_1 and θ_2
c) greater than θ_2 d) less than θ_1 .
- ⑤ Explain the characteristics of a p-n junction diode that makes it suitable for its use as a rectifier.
- ⑥ A particle of mass m and charge q describes a circular path of radius R in a magnetic field. If its mass and charge were $2m$ and $\frac{q}{2}$ respectively. Then the radius of its path would be ?